

# Question Paper

Exam Date & Time: 30-Dec-2025 (02:00 PM - 05:00 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH SEMESTER B.TECH. DEGREE, MAKEUP EXAMINATIONS - DECEMBER 2025

Human Computer Interaction [NPT\_4131]

Marks: 50

Duration: 180 mins.

### PART-A

Answer all the questions.

- 1) Construct a mid-fidelity prototype for a healthcare mobile application. Build a user flow that demonstrates navigation between key screens, and model how user feedback at this stage could refine the final high-fidelity prototype. (5)
- A)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Categorize the main interaction types in HCI (Instructing, Conversing, Manipulating, Exploring, and Responding). Distinguish their unique roles, and explain the function of multimodal interaction in enhancing user experience. (3)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- C) Analyze the relationship between SECI and IoT to determine how user-centered design principles enable effective communication between humans and connected devices. (2)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Build a short step-by-step plan outlining how you would gather qualitative data through *semi-structured interviews* for a usability study of an online learning platform. (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Apply the concept of *IRB approval* to a research scenario involving user testing of a mental health app. Explain how ethical standards and participant consent should be maintained. (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- C) Apply the concept of IRB approval to explain why a researcher studying gamified interfaces must include a *plan for anonymizing participant data* in their proposal. (2)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 3) Construct a prototyping plan for a new e-learning mobile app by outlining the sequence of steps you would take from *lo-fi sketches* to *hi-fi prototypes*. Explain how you would incorporate *user testing and stakeholder feedback* at each stage. (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Classify the tools mentioned (paper-pencil, Miro, Canva, Figma, Adobe XD) by the fidelity stage where they are primarily used. Categorize each tool by its primary function. Justify each placement (3)

briefly

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) Determine the most suitable fidelity level to demonstrate the design of a food delivery application and justify your selection briefly. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 4) Examine how user feedback influences transitions between different fidelity stages in the design process. Draw an inference about how continuous user testing enhances usability and stakeholder confidence. (5)
- A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Determine an appropriate data analysis tool (R, SPSS, or NVivo) for a study focusing on emotional responses in interface design, and justify your choice in one or two sentences. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Examine how continuous user feedback gathered through tools like Figma or Canva helps refine design elements during the transition from Mid-Fi to Hi-Fi prototypes, and provide a brief inference on the indicators that show a design is ready for stakeholder presentation. (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Construct a user-centered prototyping plan for a mobile healthcare application. Include the choice of fidelity level (Lo-Fi, Mid-Fi, Hi-Fi), the prototyping tools, and the user feedback methods you would employ at each stage. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Compare the ethical requirements for IRB-approved research with the privacy and security principles in SECI-based systems. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Identify two major ethical safeguards required during data gathering under the IRB approval process and explain why they are essential in human-centered research. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

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| Event Name | Subject |
|------------|---------|
|------------|---------|

058-MIT BLR-MAKEUP EXAM-2025-2026

NPT\_4131:Human Computer Interaction

| #     | Course Outcome | Description | Marks |
|-------|----------------|-------------|-------|
| 1     | 1              | CO 1        | 7     |
| 2     | 2              | CO 2        | 5     |
| 3     | 3              | CO 3        | 23    |
| 4     | 4              | CO 4        | 15    |
| Total |                |             | 50    |

\* **NA** - CO Not Selected

| Event Name | Subject |
|------------|---------|
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058-MIT BLR-MAKEUP EXAM-2025-2026

NPT\_4131:Human Computer Interaction

| # | Level       | No of Questions | Marks (%) |
|---|-------------|-----------------|-----------|
| 1 | Remember    | 0               | 0(0%)     |
| 2 | Understand  | 0               | 0(0%)     |
| 3 | Application | 9               | 32(64%)   |
| 4 | Analysis    | 6               | 18(36%)   |
| 5 | Create      | 0               | 0(0%)     |
| 6 | Evaluate    | 0               | 0(0%)     |
|   | Total       | 15              | 50 (100%) |